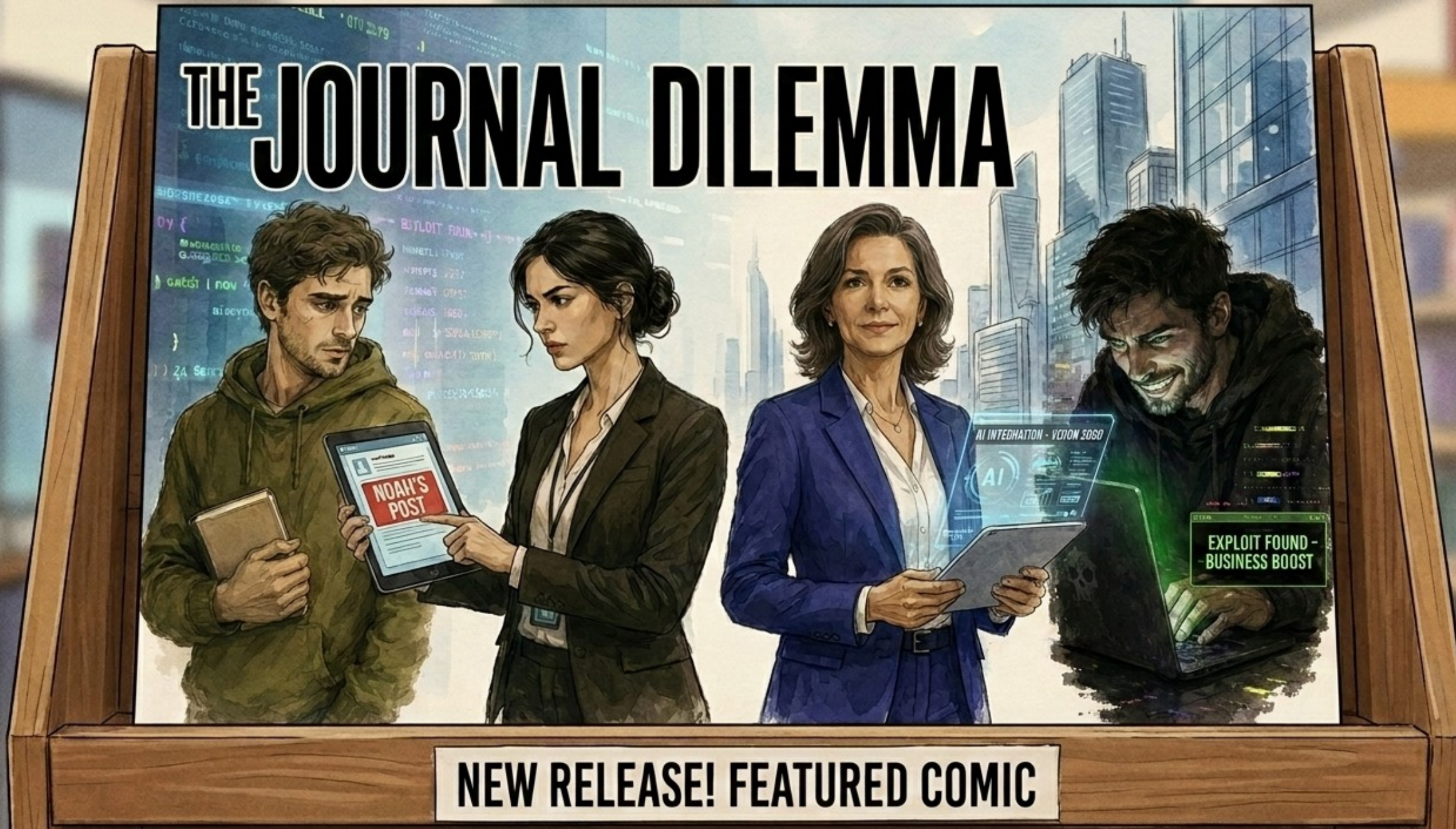


How far can you go with AI without sharing your data?

Local use | Computation on encrypted data | Hardware isolation

AFONSO OLIVEIRA — OLIVEGRADIENT.COM


THE JOURNAL DILEMMA
NOAH'S POST
AI INTEGRATION - VERSION 5050
EXPLOIT FOUND - BUSINESS BOOST
NEW RELEASE! FEATURED COMIC

Noah writes; Lily hosts



THE NAIVE BASELINE

Plaintext on the server

- **Noah's entries travel unencrypted**

  The data reaches the server in the clear.

- **Anyone with server access reads everything**

  Lily can. So can a breach, a subpoena, or an insider.

Privacy is simply broken.





Encrypt it — now you can't use it



Enter **Caroline** — the CEO who wants to ship AI features and can't.

01

PART ONE

Why not living on the edge

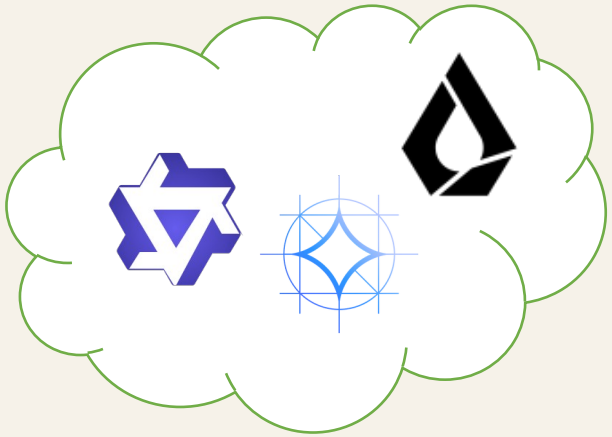
Some good solutions appear



Google AI Edge



Llama Edge



Optimized Small Models

Now the model is the thing exposed

- **Run the model on Noah's device**

  Data stays local — Noah's privacy is solved.

- **But the weights now sit on a device you don't control**

  Caroline's model can be extracted or replicated.

Enter **Mario** — who extracts weights or copies the model to rival JournX.



Just some tweaks to this and I can compete with them.
The threat to Caroline's IP.

THE CORE PROBLEM

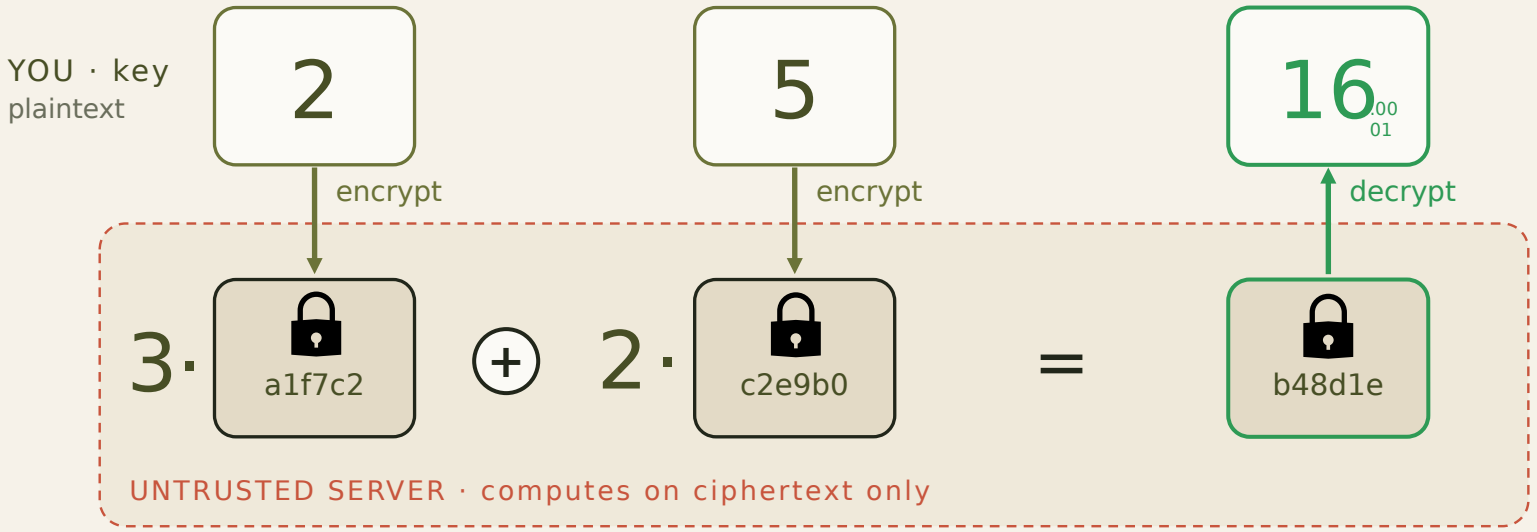
Two secrets now need protecting at once: Noah's data and Caroline's model . And for a language model, that is genuinely hard.

02

PART TWO

Computing on encrypted data

Math that survives encryption



ADD

Enc(a) + Enc(b)
≈ Enc(a + b)
two ciphertexts in, the encrypted sum out

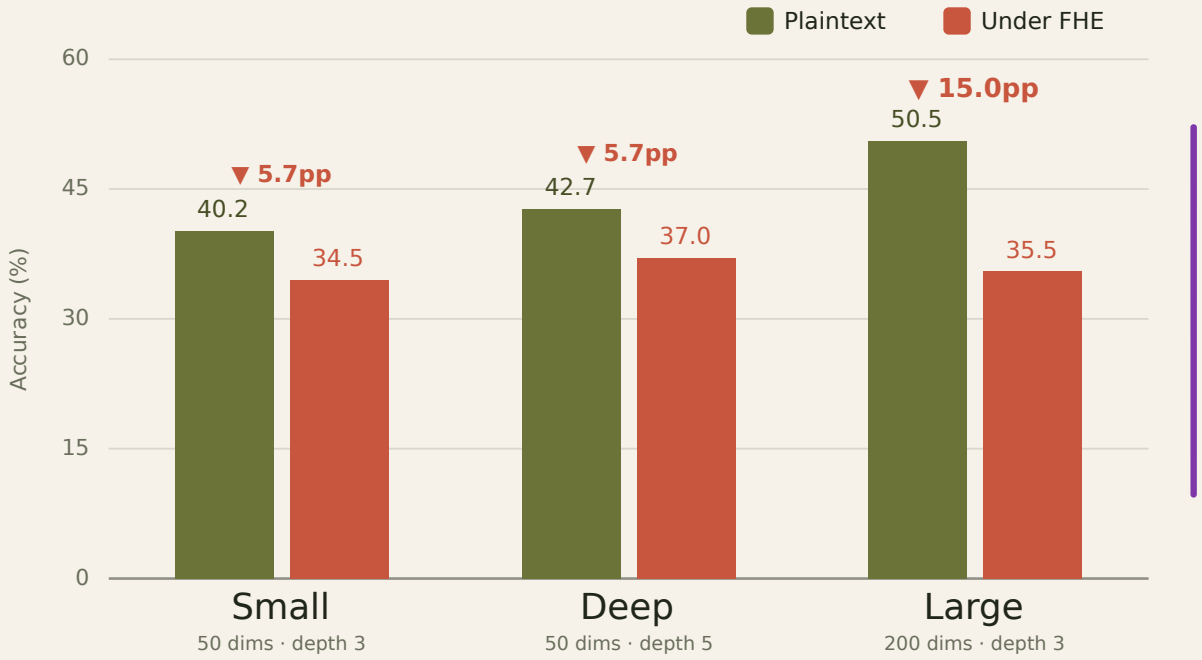
SCALE

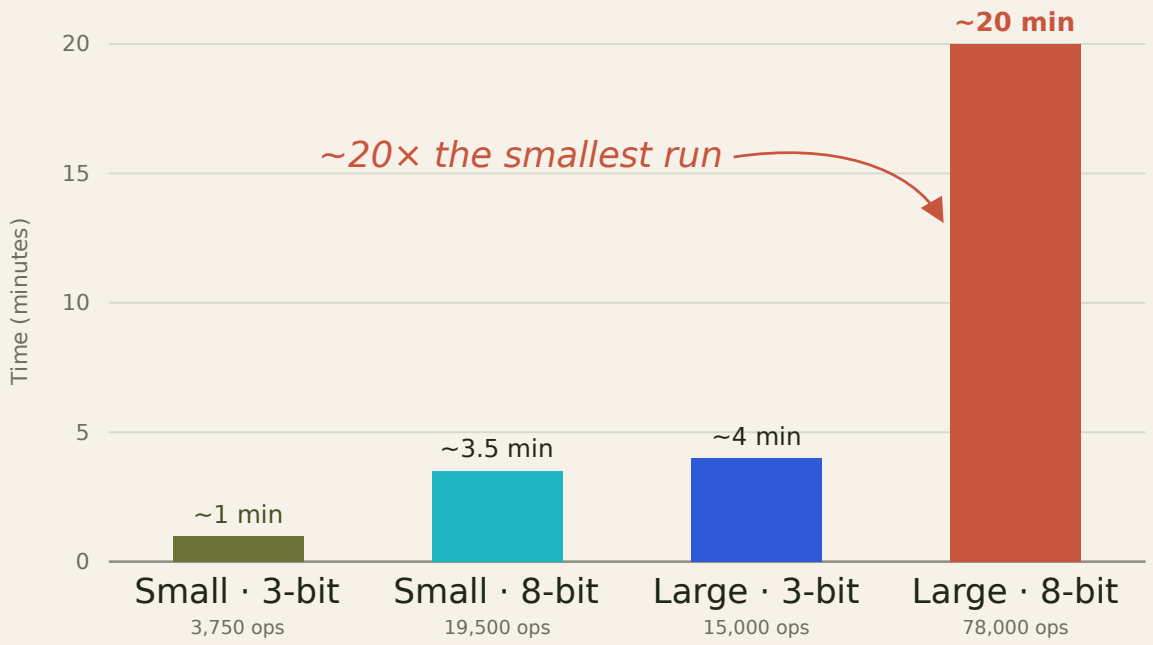
k · Enc(a)
≈ Enc(k · a)
multiply by a constant known in the clear

- **The catch**

  Every operation adds noise – requires quantization. Non-linear operations (eg. ReLU) are approximated with expensive operations.

Emotion detection on encrypted text – Results





WHY WE NEED A DIFFERENT APPROACH

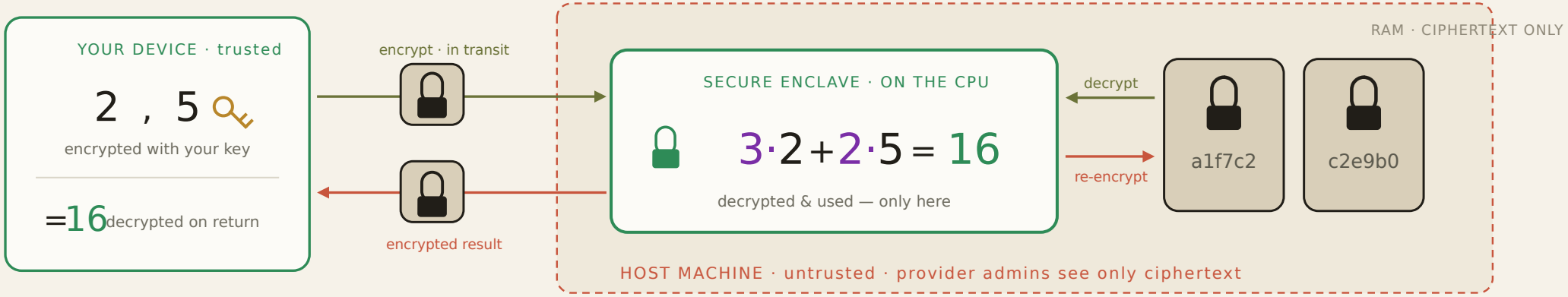
FHE could in principle hide the model too — but the depth of a language model breaks it. We need protection that doesn't grow with every operation.

03

PART THREE

Hardware-isolated enclaves

A sealed room in the hardware



- **The catch**

  Your data is decrypted inside the enclave — so you trust the chip's hardware, not just math (as in FHE).

  Everywhere else — the wire, the RAM, the host OS — it stays sealed.

Proof of exactly what's running

STEP 01

Code

You specify the exact program that should run.

→

STEP 02

Report

The TEE measures what loaded and signs a hash of it in hardware.

→

STEP 03

Verify

You check the hash matches what you expected — before trusting it.

Attestation proves the code inside matches what you specified in a TEE confidential VM.

TWO KINDS OF GUARANTEE

Where the trust actually sits

FHE

A mathematical guarantee.

Nobody can reach the plaintext — not even you, without the key. The cost is performance.

TEE

A hardware guarantee.

You verify the code, but you must trust the chip manufacturer. The reward is full speed.

WHAT YOU CAN USE TODAY

RedPill and Tinfoil AI

YOU

Verify

Check who you are talking to: TEE + model.

→

YOU

Encrypt

Send a request, encrypted end-to-end.

→

TEE

Process

The LLM runs inside the enclave.

→

BACK TO YOU

Answer

Encrypted response, only you can open.

Use it like an endpoint — just verify it first and encrypt the payload.

Caroline's corner stays unsolved



INTERNAL USE — WORKS

Caroline can process her own data privately.

For her own queries, the data stays confined. Redpill is enough.

SERVING USERS — DOESN'T

She can't serve *her* model this way.

Attestation would mean opening her model code — exposing the IP she's trying to protect.

04

PART FOUR

Augment on TEE basis

Three parties, one machine none of them own



Caroline
Private IP: weights | harness

contract

One machine
none of them controls

contract



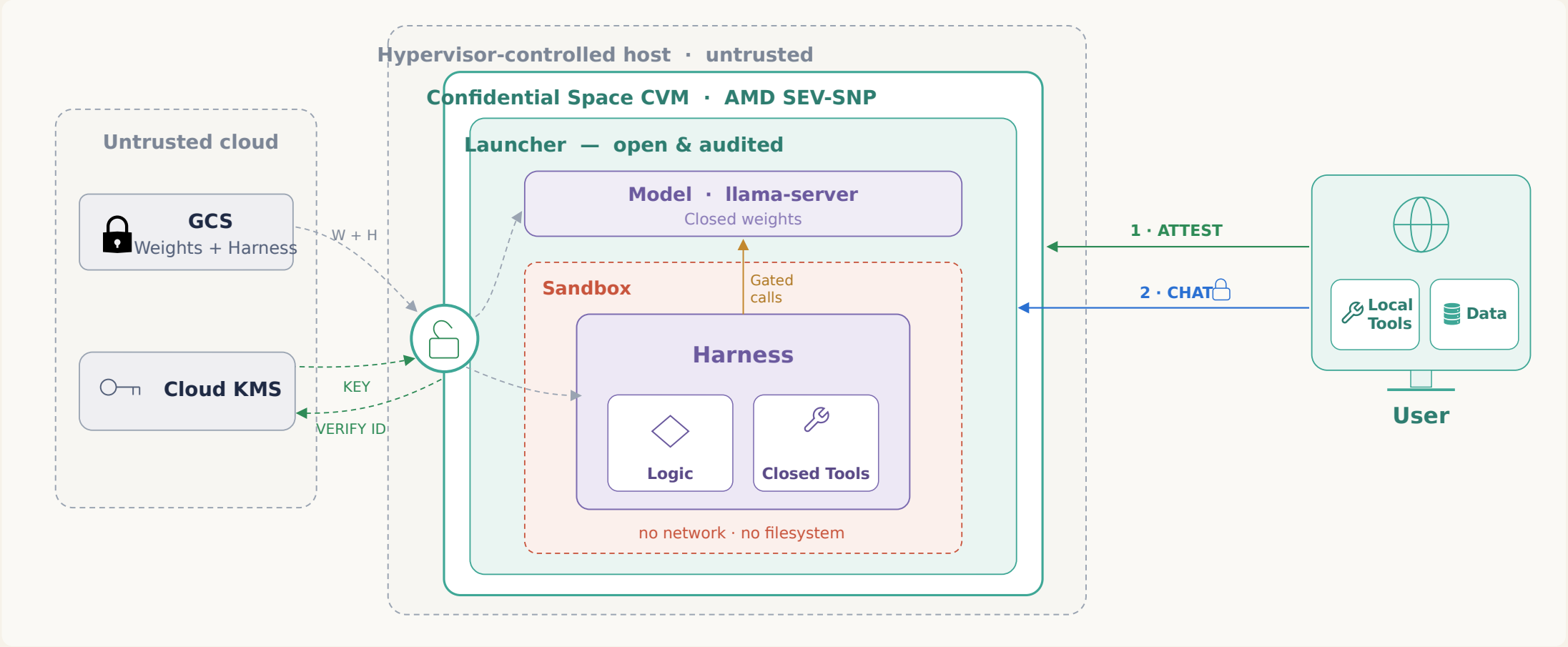
Mario
Works at the hypervisor

contract



Noah
Private journal entries

Using GCP Confidential Space



WHERE THIS LEAVES US

Room to create, but no maturity. Stay tuned!

READ MORE

olivegradient.com

THANK YOU

Afonso Oliveira